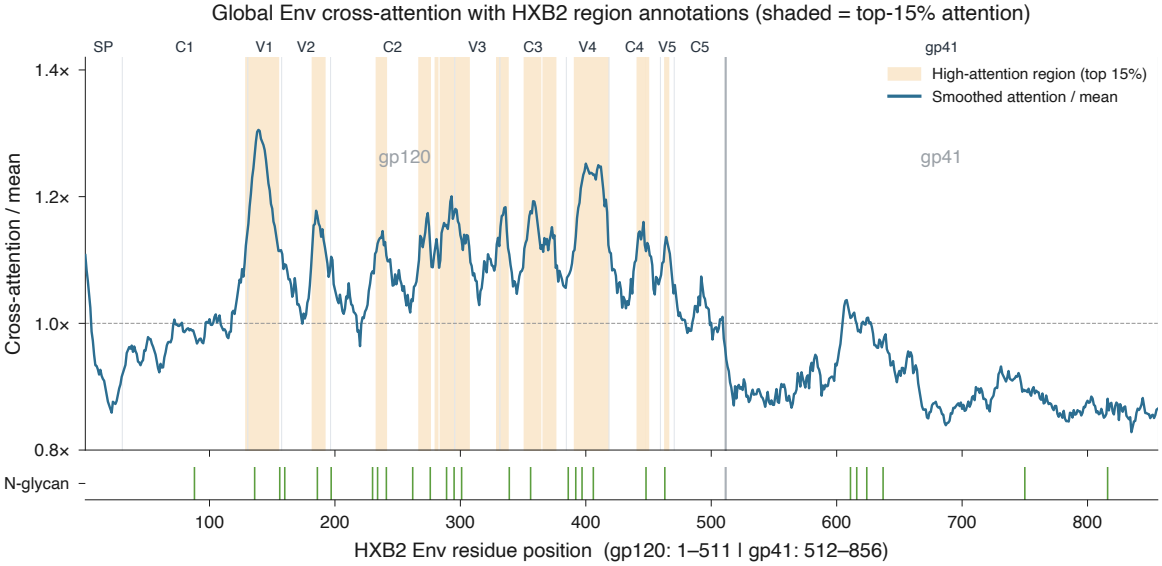
Global Env cross-attention with HXB2 region annotations (shaded = top-15% attention)
SP
C1
V1
V2
C2
V3
C3
V4
C4
V5
C5
gp41
High-attention region (top 15%)
Smoothed attention / mean
gp120
gp41
1.4×
1.2×
1.0×
0.8×
Cross-attention / mean
N-glycan
100
200
300
400
500
600
700
800
HXB2 Env residue position (gp120: 1–511 | gp41: 512–856)